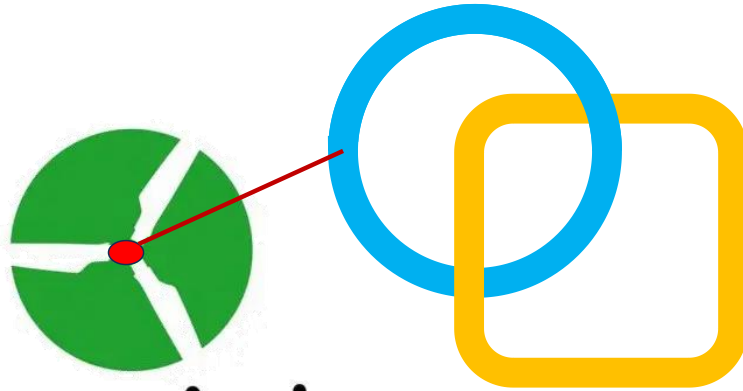




iea wind *Task 52*

LAC Summer Games 2026

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Introduction

Following the success of the Lidar-Assisted-Control (LAC) Summer Games in 2024 and 2025, this new edition builds on the lessons learned that were presented at the Wind Energy Science Conference in June 2025 [1] and at the LAC working group meeting [2]. From an educational perspective, the Summer Games 2024 proved to be a great opportunity for problem-based learning and encouraging collaboration between universities and professionals from industry. In total, 15 contributions from students and professionals across 6 countries have been submitted to three different disciplines. From the 15 contributions, 7 detailed descriptions have been uploaded to the [IEA Wind Task 52 community on Zenodo](#). The Summer Games 2025 focused more on Students and on updated scenarios and more realistic wind conditions. In total, 7 student contributions from across 3 countries have been submitted to two different disciplines. From all contributions, detailed descriptions have been uploaded to [Zenodo](#). At the same time, the experience showed how challenging it is to design LAC exercises that are realistic enough to be meaningful, yet manageable within a limited timeframe. For the Summer Games 2026, we want to continue this collaborative format with a challenge focusing on real world data.

Background

Wind characteristics are inherently dynamic, subject to sudden changes in speed, direction, and turbulence. Consequently, wind turbines are highly dynamic systems, excited by such stochastic influences and must be designed to handle these variations and disturbances. Traditional controllers react only after wind changes have impacted the wind turbine, resulting in an offset operation until the controller handles the disturbance. In contrast, lidar-assisted control concepts use preview information of the wind to anticipate changes, hence reacting before the disturbance approaches the rotor and therefore improving the operation of the turbine.

The mission of our IEA Wind Task 52 working group is to push forward the lidar-assisted control technology and simplify its application by recommended practices, open-source tools, and joined exercises. In line with our mission, we are very glad to launch our Summer Games 2026 after the successful Summer Games 2024 and 2025 to encourage students and professionals in the art of designing and deploying lidar data processing algorithms and lidar-assisted controllers.

By participating in the games, you can increase your knowledge in LAC and enjoy the design process by competing with others. Moreover, we aim to trigger your creativity in two different disciplines and with it motivate the development of new concepts. During the competition, it is intended that the participants learn the existing open-source tools and their application will result eventually in the further development of the tools. Additionally, during the games you will be able to exchange knowledge with different parties of the wind energy community. We are looking forward to strengthening your passion for advancing the vision of modern wind turbines and getting the best LAC out of you! The tools are based on previous publications [2], [4] and built up on the NREL 5 MW reference wind turbine [5]. We intend to write a publication on the results similar to [1] and integrate the learnings in our planned IEA Recommended Practices on Lidar-Assisted Control.

Our Categories

For the Summer Games we have the following categories:

1. Students (including PhD students)
2. Researchers

Participation in groups is possible. If all group members are students, the first category applies.

Evaluation and awards

In every discipline (see below) we will rank the contributions for each of the two categories mentioned above depending on the following: the contribution with the lowest Smallest Detectable Eddy Size (SDES) or the lowest load reduction ratio wins (minimization problem). The results will be shared on the IEA Task LinkedIn page, and participants will also have the opportunity to present their solution to the IEA Wind Task 52 LAC working group, which consists of a selection of academic and industry experts. Further, we will provide certificates and a trophy.

WETI Research Turbine for LAC

The measurement data used in the LAC Summer Games 2026 were recorded on a commercial Senvion 6.2M126 wind turbine at the WETI research site in Wanderup near Flensburg. The site comprises two turbines of the same type equipped with different nacelle-mounted lidar systems, enabling comparative investigations under similar inflow and operating conditions. The turbines are owned and operated by Denker & Wulf AG and used by WETI for research and development activities. The data provided here are owned by WETI and can only be used for this competition. Further use can be requested contacting the support team (see below).

The instrumented turbine is equipped with a nacelle-mounted lidar system and synchronized turbine data acquisition. This setup provides simultaneous measurements of the approaching wind field and selected turbine responses. Unlike purely synthetic datasets, the measurements reflect field conditions, including varying inflow, measurement uncertainty and periods of reduced lidar data availability.

The field measurements support the reconstruction of rotor-effective wind speed and provide inflow information for the simulation-based assessment of load-reducing control strategies.



Figure 1 Lidar-equipped 6.2M126 research wind turbine in Wanderup, Photo: M. Schedat.

Our Disciplines

In this section, you will find the different disciplines where you can participate with your proposed lidar-data processing algorithms and lidar-assisted controllers (it is also possible to just improve the controller). Regardless of your chosen category, we encourage you to submit outcomes for as many disciplines as you believe align with your capabilities. In the following subsections, each of the disciplines is presented together with the evaluation methods used for evaluation.

The 6.2M Ultramarathon – Wind Preview Quality

In the first challenge we'll test what you can get out of the lidar measurements. To accomplish this, we restrict you to the use of the lidar data only (*lineOfSightWindSpeed*, *isValid* and *beamID*). The challenge is addressing mainly lidar experts, and the intention is to introduce you to lidar-data-processing for control and to trigger new lidar-data processing (LDP) algorithms using real measurement data. The objective is to improve the LDP to reduce the smallest detectable eddy size as much as possible.

For this purpose, the real lidar measurements are handed to an LDP-algorithm to compute a preview of the rotor effective wind speed (REWS). The true REWS is estimated via an observer and provided within signal v_0 : this signal is filtered and shifted in time, so that frequencies such as the 3P caused by the turbine are damped and the lag of the estimation process is discarded via the time shift. The lidar data contains the *lineOfSightWindSpeed*, the quality flag *isValid*, indicating for every *lineOfSightWindSpeed*, if the measurement is valid (true/false) and the *beamID*, indicating from which of the 4 beams the current measurement is coming from: 1 = top-left, 2 = top-right, 3 = bottom-left, 4 = bottom-right. The lidar measurement distances have been set to 50, 65, 80, 100, 115, 130, 145, 160, 175, and 200 m and the lidar installed at 3.1 m behind the rotor and 4.1 m above the hub center. The lidar has a 4 Hz scan rate and +/- 15 deg horizontal and +/- 12.5 deg vertical angles.

The data set covers 12 h recorded on April 5, 2026 with a sampling rate of 20 Hz (864000 samples with a time step of 0.05 s). The rotor-effective wind speed v_0 ranges from 11.0 to 26.6 m/s with a mean of 18.5 m/s and a standard deviation of 2.46 m/s, so the turbine is in full load operation over practically the whole time. Over all range gates, 63.5 % of the line-of-sight measurements are valid, but the availability differs between the beams: about 74 % for the beams 1 and 2 and about 52 % for the beams 3 and 4.

The script *RunUltraMarathon.m* (or *RunUltraMarathon.py*) in the *UltraMarathon/matlab* (or *UltraMarathon/python*) folder (within the “SummerGames2026” branch) provides the baseline smallest detectable eddy size using the LDP-algorithm *LDP_v3*. The used baseline lidar data processing *LDP_v3.m* (or *LDP_v3.py*) is based on the Summer Games 2025, using only one gate and ignoring measurements with a false *isValid* flag, causing the “blade ghost effect” [6]. The smallest detectable eddy size is computed via the coherence between signals v_0 and v_{0L} referring to the estimate of v_0 via the LDP-algorithm. The coherence over wavenumber k is illustrated in Figure 2. The evaluation is done over the whole period of 12h.

The evaluation criterion for this discipline is the smallest detectable eddy size (SDES) defined as:

$$\text{SDES} = \frac{2\pi}{k_{0.5}D},$$

where $k_{0.5}$ is wave number where the magnitude squared coherence reaches 0.5 ($\gamma_{RL}^2 = 0.5$) and D is the rotor diameter of the turbine. The result for the baseline *LDP_v3* algorithm is: 1.738 D .

The following rules apply:

1. Your LDP-algorithms can only use lidar signals (*lineOfSightWindSpeed*, *isValid*, *beamID*). No future information from the turbine or the ground truth (v_0) itself can be used within your LDP-Algorithm. The algorithm can only use current and past data for the current prediction.
2. New lidar data processing ideas should be implemented with the same interface as the function *LDP_v3.m* (or *LDP_v3.py*) in the loop over each time step.

Following references could help you:

- Real-time wind field reconstruction from LiDAR measurements using a dynamic wind model and state estimation [7]
- Multi-Distance Lidar Data Processing [8]
- Wind field reconstruction based on PINNs [9]
- Kalman Filter implementation for wind field reconstruction [10]
- Improved Wind and Lidar Models [11]
- Publications from previous summer games on [Zenodo](#).

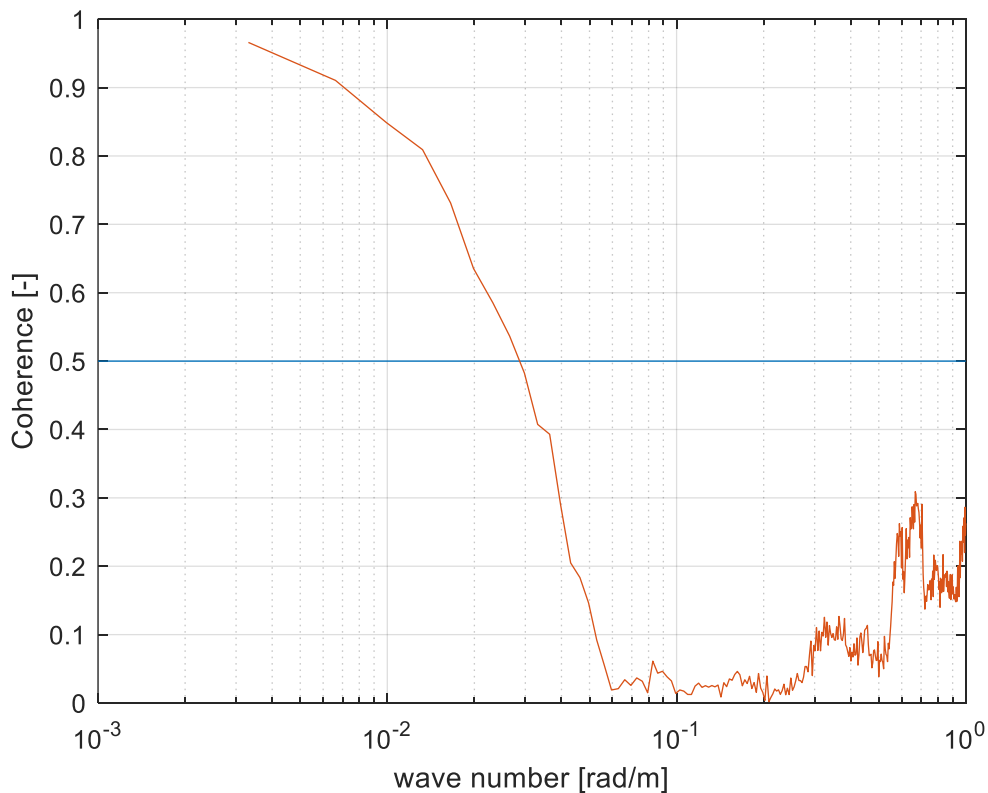


Figure 2: Coherence between REWS (v_0) and lidar estimated REWS (v_{0L}).

The 6.2M Ultramarathon – Load Reduction

In the second challenge, we'll test how your LDP-algorithm can be used for control and how much your control performance is improved for the whole 12-hour data set. The challenge is mostly addressing control experts, and the intention is to introduce you to the use of lidar data for wind-turbine-control and to trigger new control ideas. The objective is to reduce the damage equivalent loads (DEL) of the tower (in fore-aft direction) in comparison to standard feedback controller. You don't have to use the lidar signal, one could also just improve the feedback controller.

The script *RunUltraMarathon.m* (*RunUltraMarathon.py*) is first calculating the DEL for the standard feedback (FB) controller as reference. Then a simple collective pitch feedforward (FF) controller is added to the feedback controller as a simple LAC example and starting point for development. The simulation is done with the Simplified Low Order Wind turbine model (SLOW) of the NREL 5 MW turbine. The NREL 5 MW reference turbine is used here because it is close to the Senvion 6.2M126 on which the data has been measured: it has the same rotor diameter of 126 m and comparable further properties. It is a 2 degree of freedom low order model, simulating the tower fore-aft motion and the rotation of the rotor. In addition, a pitch actuator is added, modelled as a second order system with a natural frequency of 1 Hz, a damping of 0.7 and a pitch rate limit of 8 deg/s. The different effects both controllers have on the pitch, generator torque, rotor speed and tower base bending moments are shown in Figure 3.

The evaluation criterion for this discipline is the reduction of the DEL in comparison to the DEL using the standard feedback controller defined as:

$$J = \frac{DEL_{LA}}{DEL_{FB}},$$

where $(.)_{LA}$ indicates the lidar assisted controller (or any other new controller) and $(.)_{FB}$ the baseline feedback controller respectively. This ratio is printed by the scripts as Cost. In addition to the loads the following restrictions for the controller are set:

1. The energy yield is not allowed to drop by more than 1 kWh (over the 12 h data set) below the value of the FB case.
2. The maximum rotational speed of the FB case is not allowed to be exceeded.
3. The standard deviation of the electrical power of the FB case is not allowed to be exceeded.
4. The pitch travel of the FB case is not allowed to be exceeded.

All restrictions are evaluated over the full 12 h and compared to the FB case: the maximum rotational speed as the maximum of the generator speed (first measurement in y), the power quality as the standard deviation of the electrical power (fourth measurement in y), and the pitch travel as the sum of the absolute increments of the actual (not the commanded) pitch angle (second measurement in y). A contribution which violates one of these restrictions is not ranked in this discipline.

The cost for the baseline LAC against the pure feedback is 94.96%.

The following rules apply:

1. You can replace the feedback and feedforward controller (in the section: "simulation lidar-assisted: please only adjust code between >>> <<<!") and use your own feedback and feedforward controller, model predictive control etc. adjusting collective pitch angle and generator torque.

2. It is not allowed to change the model itself or the FB simulation.
3. Only measurable states of the models such as generator speed, pitch angle, tower top acceleration and electrical power can be used.
4. The true rotor-effective wind speed v_0 must not be used in the lidar-assisted simulation, neither directly nor delayed. It is the ground truth, serves only as disturbance input of the model and is used for the evaluation.

Following references could help you:

- Collective pitch feedforward control [3], [8]
- Multi-variable feedforward control [12]
- Model predictive control [13]
- Nonlinear output regulation [14]
- Publications from previous summer games on [Zenodo](#).

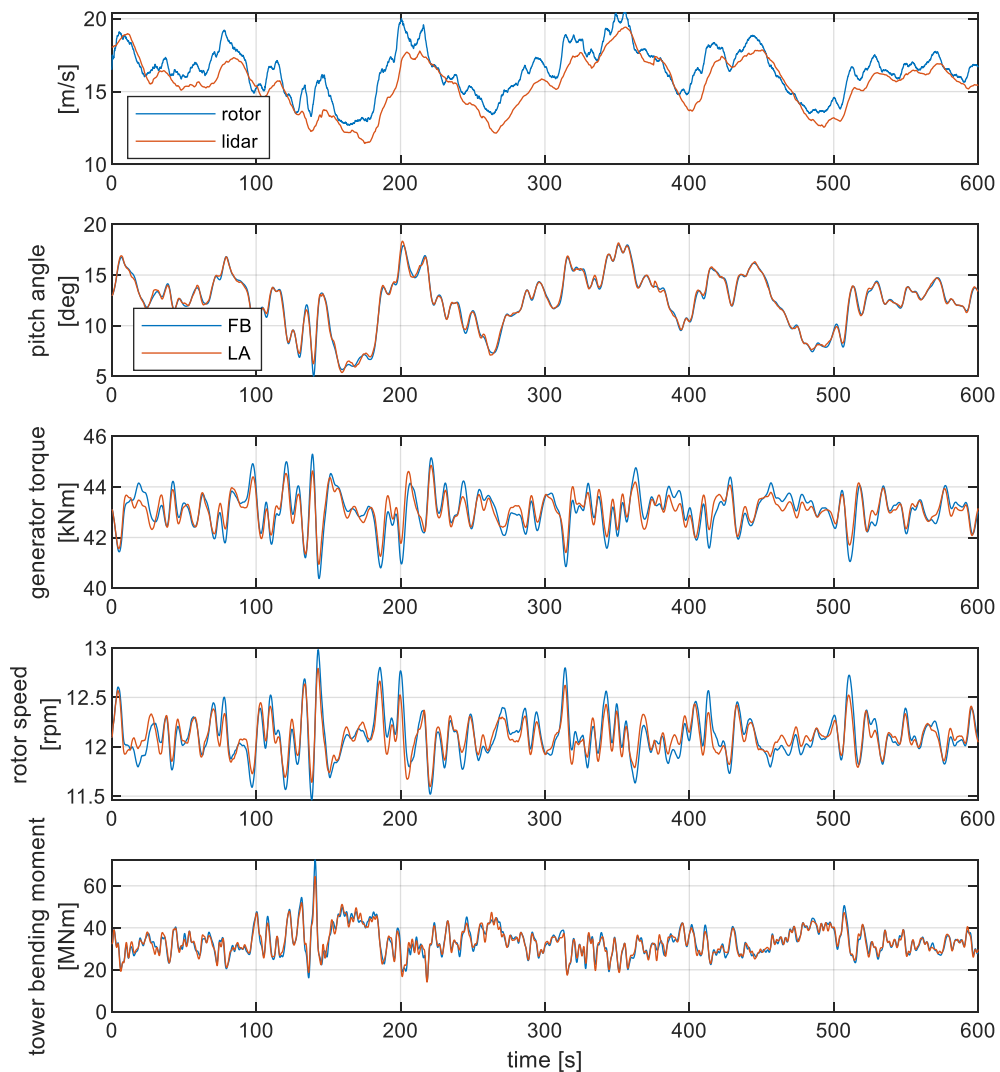


Figure 3: Controller comparison sample (first 600s) of the 12h simulation using the SLOW model and real measurement data.

Our AI Guidelines

AI-assisted writing and coding is generally acceptable, subject to (a) respecting copyright, (b) declaring that AI was used, (c) document which AI model/system has been used and describe in details how it was used (e.g. which code/text sections were partly/fully generated by AI), and (d) the author remaining fully responsible for the codes generated and the submitted documents.

Our timeline

The Summer Games 2026 will start on August 1, 2026, and will run until February 28, 2027 (end of winter semester of most universities).

What should be provided:

1. Please provide the result file including the x_{LA} , u_{LA} , y_{LA} and v_{OL} time series per email to the support team (see below). Please use a MATLAB *.mat or a NumPy *.npz file with exactly these variable names and the full sampling rate of 20 Hz and report the smallest detectable eddy size and the cost printed by the script.
2. If possible: In addition to the files above, please provide DLLs and other files which are necessary to run simulations or lidar data processing algorithms (via a Git fork). Please check with someone independently to ensure that the scripts run and state clearly which script produces the submitted files.
3. Please provide a document describing the methodology and results using the abstract template provided by the WESC conference, see <https://www.wesc2027.eu>.

We will then evaluate the results together in our working group meetings and announce the rankings in the two disciplines and two categories. Additionally, new data from the same turbine and lidar will be provided to check the robustness of the algorithms (not affecting the ranking).

Any changes to the rules or submission dates can be discussed in the discussion forum <https://github.com/IEAWindTask52/LidarAssistedControl/discussions>.

How to get started

To get started with our Lidar-Assisted Control open-source tools, you need to clone the [GitHub repository](#) to your local computer using either Git or GitHub Desktop. Please make sure you change to the “SummerGames2026” branch. You can use either Matlab or Python.

To get started with the Matlab or Python script you need to navigate to the *UltraMarathon/matlab* or *python* folder and then you can reproduce the results above via *RunUltraMarathon.m* (or *RunUltraMarathon.py*) script. The script is using functions that are provided in the *functions* folder, and the data is loaded from the *data* folder (UltraMarathon/data), which is shared by the Matlab and the Python version. In Matlab, the functions and the data folder are added to your path; in Python, the script adds the functions folder itself and loads the data via the relative path *../data*.

To get started with Python scripts unlike Matlab users need to install various modules like numpy, scipy, h5py, matplotlib and rainflow. To make it simpler we have provided a *setup_python.py* script that installs all the necessary modules in your environment.

Support

If you have questions regarding the Summer Games in general or regarding the code which might be interesting for others as well, please use our Forum on [Github](#).

If you require further support, please don't hesitate to contact our support team via email:

- Felix Lehmann felix.lehmann@hs-flensburg.de
- Simon Weich simon.weich@hs-flensburg.de
- David Schlipf david.schlipf@hs-flensburg.de

References

- [1] Schlipf, D., Weich, S., Raach, S., and Guo. F.: Evaluation and Lessons Learned from the Lidar-Assisted Control Summer Games 2024. Presentation at Wind Energy Science Conference. DOI: [10.5281/zenodo.16105260](#), 2025.
- [2] Schlipf, D., Guo, F., van den Broek, M., Weich, S. and Preuschoff, J.: Evaluation of the Lidar-Assisted Control Summer Games 2025 and Update on LAC Open Source Tools. Zenodo. DOI: [10.5281/zenodo.20625931](#), 2026.
- [3] Guo, F., Schlipf, D., and Cheng, P. W.: Evaluation of lidar-assisted wind turbine control under various turbulence characteristics, Wind Energ. Sci., 8, 149–171, DOI: [10.5194/wes-8-149-2023](#), 2023.
- [4] Schlipf, D., Guo, F., Raach, S., and Lemmer, F.: A Tutorial on Lidar-Assisted Control for Floating Offshore Wind Turbines, American Control Conference, DOI: [10.23919/ACC55779.2023.10156419](#), 2023.
- [5] Jonkman, J., Butterfield, S., Musial, W. and Scott, G.. Definition of a 5-MW Reference Wind Turbine for Offshore System Development. Tech. rep. TP-500-38060. NREL, DOI: [10.2172/947422](#), 2009.
- [6] Schlipf, D., Lemmer, F., and Thomas, F.: Realization of wind field reconstruction for real-time monitoring and LIDAR-assisted control. Presentation at the 5th International Conference on Offshore Renewable Energy. DOI: [10.5281/zenodo.5608684](#), 2021.
- [7] Towers, P., and Jones, B. L. (2016) Real-time wind field reconstruction from LiDAR measurements using a dynamic wind model and state estimation. Wind Energ., 19: 133–150. DOI: [10.1002/we.1824](#).
- [8] Schlipf, D.: Lidar-Assisted Control Concepts for Wind Turbines, Ph.D. dissertation, University of Stuttgart, 2016, Dr. Hut Verlag, ISBN [978-3-8439-2518-1](#), DOI: [10.18419/opus-8796](#).
- [9] Zhang, J., and Zhao, X. (2021) Spatiotemporal wind field prediction based on physics-informed deep learning and LIDAR measurements, Applied Energy, 288, 116641. DOI: [10.1016/j.apenergy.2021.116641](#).
- [10] Correa, S., Draper, M., & Schlipf, D. (2026). Wind field reconstruction for lidar-assisted control of wind turbines. Journal of Physics: Conference Series, 3224(5), Article 052038. DOI: [10.1088/1742-6596/3224/5/052038](#).
- [11] Guo, F., Improved modeling of lidar wind preview for wind turbine control, Ph.D. dissertation, University of Stuttgart, DOI: [10.18419/opus-12880](#).
- [12] Schlipf, D.: Prospects of Multivariable Feedforward Control of Wind Turbines Using Lidar, American Control Conference, Boston, USA, July 2016 (invited), DOI: [10.1109/ACC.2016.7525112](#), [Link](#).
- [13] Loew, S. and Bottasso, C. L.: Lidar-assisted model predictive control of wind turbine fatigue via online rainflow counting considering stress history, Wind Energ. Sci., 7, 1605–1625, DOI: [10.5194/wes-7-1605-2022](#), 2022.
- [14] Moldenhauer, R. H. and Schmid, R.: Lidar-assisted nonlinear output regulation of wind turbines for fatigue load reduction, Wind Energ. Sci., 10, 1907–1928, DOI: [10.5194/wes-10-1907-2025](#), 2025.